EXPERIMENT 1:

Characteristics Study of Side Band Power and Frequencies in AM Using MATLAB

Aim: To generate an Amplitude Modulated (AM) signal using MATLAB and study its sideband frequencies and power. To calculate the carrier power, upper and lower sideband power, and total AM power.

Soft ware Required: MATLAB

Program:

```
clc;

clear;

close all;% Characteristics Study of Sideband Power and Frequencies in AM

% Step 1: Define parameters

Ac = 10;         % Carrier amplitude (V)

Am = 5;          % Message signal amplitude (V)

fc = 1000;       % Carrier frequency (Hz)

fm = 100;        % Message frequency (Hz)

Fs = 10000;      % Sampling frequency (Hz)

t = 0:1/Fs:0.05;  % Time vector

% Step 2: Calculate modulation index

m = Am / Ac;

fprintf('AMPLITUDE MODULATION ANALYSIS\n');

fprintf('----------------------------\n');

fprintf('Carrier amplitude = %.2f V\n', Ac);

fprintf('Message amplitude = %.2f V\n', Am);
```

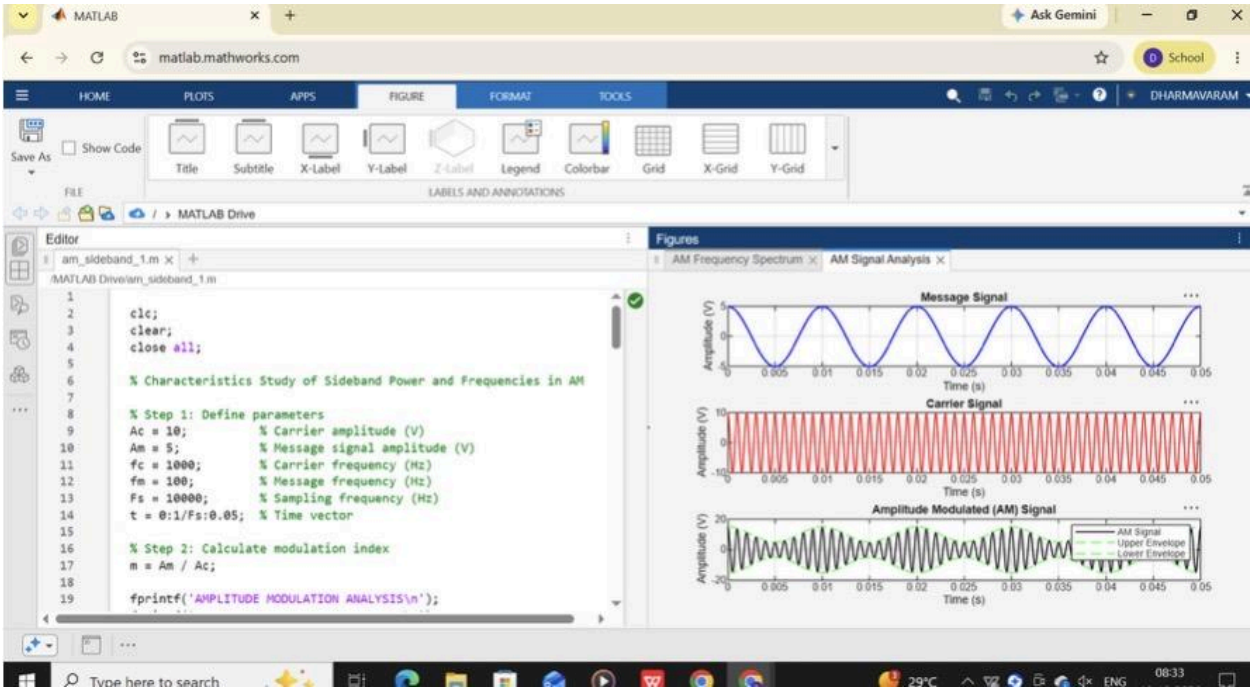
```
fprintf('Carrier frequency = %.2f Hz\n', fc);
fprintf('Message frequency = %.2f Hz\n', fm);
fprintf('Modulation index = %.2f (%.1f%%)\n', m, m*100);
% Step 3: Generate carrier and message signals
carrier = Ac * cos(2*pi*fc*t);
message = Am * cos(2*pi*fm*t);
% Step 4: Generate AM signal
am_signal = Ac * (1 + m*cos(2*pi*fm*t)) ...
        .* cos(2*pi*fc*t);
% Step 5: Calculate sideband frequencies
f_upper = fc + fm;
f_lower = fc - fm;
fprintf('\nFREQUENCY ANALYSIS\n');
fprintf('------------------\n');
fprintf('Lower sideband frequency = %.2f Hz\n', f_lower);
fprintf('Carrier frequency = %.2f Hz\n', fc);
fprintf('Upper sideband frequency = %.2f Hz\n', f_upper);
fprintf('Bandwidth = %.2f Hz\n', f_upper-f_lower);
% Step 6: Calculate power (assuming 1-ohm load)
Pc = Ac^2 / 2;              % Carrier power
Psb_each = (m^2 * Ac^2)/8;   % Power in each sideband
Psb_total = 2 * Psb_each;   % Total sideband power
Ptotal = Pc + Psb_total;    % Total AM power
```

```
fprintf('\nPOWER ANALYSIS (1-ohm load)\n');
fprintf('--------------------------\n');
fprintf('Carrier power = %.2f W\n', Pc);
fprintf('Lower sideband power = %.2f W\n', Psb_each);
fprintf('Upper sideband power = %.2f W\n', Psb_each);
fprintf('Total sideband power = %.2f W\n', Psb_total);
fprintf('Total AM power = %.2f W\n', Ptotal);
% Step 7: Plot time-domain signals
figure('Name','AM Signal Analysis','NumberTitle','off');
subplot(3,1,1);
plot(t, message, 'b', 'LineWidth', 1.2);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(3,1,2);
plot(t, carrier, 'r', 'LineWidth', 1.0);
grid on;
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(3,1,3);
plot(t, am_signal, 'k', 'LineWidth', 1.0);
```

```
hold on;
plot(t, Ac*(1+m*cos(2*pi*fm*t)), 'g--');
plot(t, -Ac*(1+m*cos(2*pi*fm*t)), 'g--');
hold off;
grid on;
title('Amplitude Modulated (AM) Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
legend('AM Signal','Upper Envelope','Lower Envelope');
% Step 8: Frequency spectrum using FFT
N = length(am_signal);
Y = fft(am_signal);
P2 = abs(Y/N);
P1 = P2(1:floor(N/2)+1);
P1(2:end-1) = 2*P1(2:end-1);
f = Fs*(0:floor(N/2))/N;
figure('Name','AM Frequency Spectrum','NumberTitle','off');
plot(f, P1, 'b', 'LineWidth', 1.3);
grid on;
title('Frequency Spectrum of AM Signal');
xlabel('Frequency (Hz)');
ylabel('Amplitude (V)');
xlim([0 1500]);
```

```
% Step 9: Display theoretical frequency components
hold on;
stem([f_lower fc f_upper], [Ac*m/2 Ac Ac*m/2], ...
   'r', 'filled');
hold off;
legend('FFT Spectrum','Theoretical Components');
```

Output:



Result:

The AM signal was successfully generated and its frequency spectrum was analyzed using MATLAB. The carrier frequency, sideband frequencies, and corresponding power values were calculated and verified.